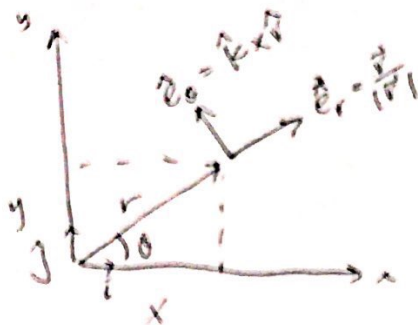


MAE 4730
Lecture 8
9/12/18

Today:
1) Polar coordinates
2) ~~Conservation~~ Angular momentum + Energy

Matlab
root finding
optimization
(fmincon)

Polar coordinates



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \begin{cases} \text{atan2}(y, x) & \checkmark \\ \text{atan}(y/x) & \times \end{cases}$$

$$\vec{r} = x\hat{i} + y\hat{j} = r\hat{e}_r$$

$$\vec{v} = \dot{x}\hat{i} + \dot{y}\hat{j} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\vec{a} = \ddot{x}\hat{i} + \ddot{y}\hat{j} =$$

$$\vec{v} = \frac{d}{dt} r \hat{e}_r = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$\dot{\hat{e}}_r = (\cos\theta\hat{i} + \sin\theta\hat{j})$$

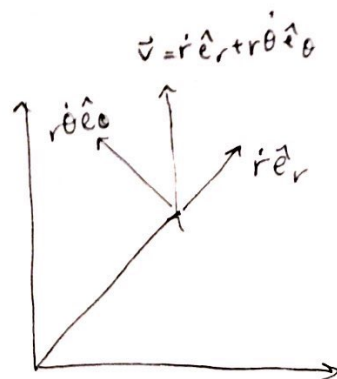
$$= -\sin\theta\dot{\theta}\hat{i} + \cos\theta\dot{\theta}\hat{j}$$

$$\boxed{\dot{\hat{e}}_r = \dot{\theta}\hat{e}_\theta}$$

$$\dot{\hat{e}}_\theta = (-\sin\theta\hat{i} + \cos\theta\hat{j})$$

$$= -\cos\theta\dot{\theta}\hat{i} - \sin\theta\dot{\theta}\hat{j}$$

$$\boxed{\dot{\hat{e}}_\theta = -\dot{\theta}\hat{e}_r}$$



$$\vec{v} = \frac{d}{dt} \vec{r}$$

$$= \frac{d}{dt} r \hat{e}_r$$

$$= \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

$$\vec{a} = \frac{d}{dt} \vec{v}$$

$$= \frac{d}{dt} (\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta)$$

$$= (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

LMB: $\vec{F} = m\vec{a}$

central force: $f(r)\hat{e}_r = m[(\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta]$

$$\boxed{\begin{aligned} \ddot{r} - r\dot{\theta}^2 &= \frac{f}{m} \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} &= 0 \end{aligned}}$$

Multiparticle Systems

L.M.B: $\sum_j \vec{F}_j^{\text{ext}} = \sum_{\text{all masses}} m_i \vec{a}_i$
 all ext forces $= M_{\text{tot}} \vec{a}_G$

$\vec{a}_G = \frac{d^2}{dt^2} \vec{r}_G$
 $\vec{r}_G = \frac{\sum m_i \vec{r}_i}{M_{\text{tot}}}$ - average position

AMB:

$\sum \vec{r}_{i/c} \times \vec{F}_i^{\text{ext}} = \sum_{\text{all masses}} \vec{r}_{i/c} \times m_i \vec{a}_i$
 ext forces

For any pt. C

$= M_{\text{tot}} \vec{r}_{G/c} \times \vec{a}_{G/c} + \sum_{\text{all particles}} \vec{r}_{i/c} \times m_i \underbrace{\vec{a}_{i/c}}_{\vec{a}_{i/c} - \vec{a}_{G/c}}$
 $\vec{H}_G$ $\vec{H}_{i/c}$

LMB:

$\vec{F}_{\text{tot}} = M_{\text{tot}} \vec{a}_G$
 $= \frac{d}{dt} M_{\text{tot}} \vec{v}_G$
 $= \frac{d}{dt} \vec{L} = \dot{\vec{L}}$

$\vec{F} = \dot{\vec{L}}$

AMB:

$\sum \vec{M}_{i/c}^{\text{ext}} = \dot{\vec{H}}_{i/c} = \dot{\vec{H}}_{G/c} + \dot{\vec{H}}_{i/G}$
 any pt. $\vec{r}_{G/c} \times M_{\text{tot}} \vec{a}_{G/c}$ $\sum \vec{r}_{i/G} \times m_i \vec{a}_{i/G}$

$\sum \vec{M}_{i/c} = \dot{\vec{H}}_{i/c}$ what is $\dot{\vec{H}}$?
 such that $\frac{d}{dt} \vec{H}_{i/c} = \sum \vec{M}_{i/c}^{\text{ext}}$

candidate definitions of $\vec{H}_{i/c}$

$\sum \vec{r}_{i/c} \times m_i \vec{v}_{i/c}$
 $\vec{r}_{i/c}$ $\vec{v}_{i/c}$
 where c' is a fixed point
 not a velocity

4 candidate definitions of angular momentum special points C =
 fixed point in $\mathcal{F}$
 const. velocity pt
 G
 a point where acceleration is parallel to $\vec{v}_G$